Sources and Extraction Planning

Data extraction is the controlled collection of information from files, databases, services, or instruments. Before collecting anything, define the question and identify the minimum fields and time range needed. Taking an entire source without purpose increases cost, privacy exposure, and the chance of handling irrelevant data.

A source should be appropriate, lawful, sufficiently complete, and trustworthy for the intended analysis. Record who created it, how it was collected, when it was updated, and any known restrictions. Data quality begins with choosing a source whose meaning and coverage are understood.

Key points

- Common sources include CSV files, spreadsheets, JSON, relational tables, and documented APIs.
- Check license, access permissions, terms, and privacy requirements before retrieval.
- Record source name, retrieval time, version, and intended use.

Files and Structured Formats

CSV stores rows and columns as text, so dates, numbers, separators, and missing-value markers may require explicit interpretation. JSON represents nested objects and arrays and may need normalization before it becomes a flat table. Spreadsheets can contain formatting, formulas, merged cells, and multiple sheets that complicate automated use.

Inspect a small sample before loading a large file. Confirm encoding, delimiter, header row, field names, and expected record count. Preserve the original file and write cleaned outputs separately so the extraction can be repeated and audited.

Key points

- Validate column names, data types, row counts, and representative values.
- Treat identifiers as text when arithmetic on them has no meaning.
- Record parsing options and how missing values are represented.

Databases and SQL

Relational databases store related information in tables, with keys linking records. SQL lets analysts select columns and filter rows close to where data are stored. Queries should request only the needed fields and records; this reduces transfer time and helps protect unrelated data.

A basic query uses SELECT to choose columns, FROM to name a table, WHERE to filter, ORDER BY to sort, and LIMIT to inspect a sample. Joins combine related tables, but incorrect join keys can duplicate or drop records. Check row counts before and after joins.

Key points

- Use explicit column names rather than SELECT * for reusable analysis.
- Check key uniqueness and expected join relationships.
- Parameterize user-supplied values in application queries; do not build unsafe SQL by string concatenation.

PRESENTATION MATTER · PAGE 4 OF 6

APIs and Reliable Retrieval

An API exposes data through documented endpoints. Requests may include query parameters for dates, categories, or page numbers and commonly return JSON. A successful response should be checked for status, expected fields, and useful metadata—not assumed to be valid merely because the request completed.

Many APIs limit request frequency or divide results into pages. Follow documentation, respect rate limits, handle temporary failures, and avoid collecting data outside the intended permission. Store credentials in protected environment settings rather than notebooks, browser code, or public repositories.

Key points

- Inspect response status, schema, pagination, and update timestamps.
- Use small requests and follow the provider's documented limits.
- Never expose access tokens, passwords, or private keys.

PRESENTATION MATTER · PAGE 5 OF 6

Ingestion Checks and Provenance

Ingestion checks determine whether retrieved data are complete and plausible. Compare expected and actual row counts, look for missing required fields, validate data types and ranges, and inspect duplicate keys. A successful download does not guarantee that the content is current, complete, or semantically correct.

Provenance records where data came from and what happened to them. A reproducible extraction log can include a source URL or database, extraction time, query or parameters, software version, number of records, and transformations. These details help another analyst recreate the dataset and diagnose changes.

Key points

- Check schema, null counts, duplicates, and date coverage.
- Keep an immutable raw copy when permitted.
- Document exclusions, filters, and failed or partial retrievals.

PRESENTATION MATTER · PAGE 6 OF 6

Practical Extraction Project

Build a small extraction notebook using an openly licensed CSV or a documented public API. State the research question and why each selected field is needed. Load or retrieve a limited sample, inspect its schema, and apply a filter that can be explained and repeated.

Finish with a short extraction report containing the source, permission or license, retrieval time, parameters, record count, quality checks, and limitations. A second person should be able to reproduce the same steps without receiving private credentials.

Key points

- Deliver: reproducible notebook, source citation, extraction log, and schema checks.
- Review: Why do API pagination and database filters matter? Which metadata help reproduce a dataset?
- Confirm that the collection method follows permission, privacy, and provider requirements.